

INCHEON, Korea, Republic of

WMO: 471120

Lat: 37.4778N

Lon: 126.6250E

Elev: 69

StdP: 100.50

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.4	-8.4	-20.4	0.6	-8.2	-18.5	0.7	-6.7	11.2	-5.8	9.7	-3.8	4.2	0	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.4	31.6	25.3	30.2	24.7	29.0	24.0	27.0	29.1	26.1	28.6	25.3	27.8	3.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.5	22.3	28.0	25.5	20.9	27.5	24.6	19.7	26.9	85.8	29.0	81.8	28.6	78.2	27.9	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.4	6.4	-12.6	33.3	2.4	1.5	-14.3	34.4	-15.7	35.3	-17.1	36.1	-18.8	37.2		
			-14.0	27.3	2.1	1.5	-15.5	28.3	-16.7	29.2	-17.9	30.0	-19.4	31.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.9	-1.5	0.9	5.9	11.9	17.2	21.7	24.8	26.0	21.9	15.8	8.1	0.8
	DBStd	9.98	4.01	3.81	3.31	3.35	2.57	1.98	2.18	2.32	2.38	3.33	4.31	4.33
	HDD10.0	1134	356	255	133	17	0	0	0	0	0	4	84	284
	HDD18.3	2678	614	488	386	194	53	2	0	3	89	307	542	
	CDD10.0	2177	0	1	5	75	224	350	458	497	356	183	28	1
	CDD18.3	678	0	0	0	2	18	102	200	238	109	10	0	0
	CDH23.3	4290	0	0	0	9	68	379	1328	2044	448	14	0	0
	CDH26.7	1008	0	0	0	2	5	48	313	587	54	0	0	0
Wind	WSAvg	2.8	3.2	3.3	3.6	3.4	2.8	2.3	2.5	2.4	2.1	2.3	2.9	3.2
Precipitation	PrecAvg	1173	18	23	39	77	87	113	294	256	139	51	52	22
	PrecMax	2010	75	90	110	354	286	439	864	761	540	247	167	67
	PrecMin	651	2	0	3	7	7	5	79	27	0	2	4	1
	PrecStd	288	16	21	25	63	59	89	148	165	127	48	34	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	13.0	16.8	24.4	27.2	29.7	33.6	33.9	29.8	25.0	19.2	12.1
		MCWB	6.7	8.1	11.1	14.9	17.3	20.2	24.9	26.1	22.4	18.2	14.4	8.0
	2%	DB	6.9	9.6	14.4	21.0	25.2	28.0	31.0	32.2	28.2	23.1	17.2	10.0
		MCWB	4.2	5.7	9.4	13.4	16.7	20.1	25.1	25.8	21.6	17.5	13.2	6.8
	5%	DB	5.4	7.7	12.4	19.0	23.5	26.7	29.6	31.0	27.0	21.7	15.6	8.4
		MCWB	2.9	4.2	7.9	12.2	16.2	20.1	24.6	25.4	21.1	16.7	11.9	5.7
	10%	DB	4.0	6.1	10.7	17.3	21.9	25.5	28.3	29.7	25.8	20.4	14.3	7.0
		MCWB	1.7	3.3	6.8	11.4	15.7	19.9	24.2	24.9	20.4	16.0	11.1	4.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	10.3	13.1	17.2	20.0	23.7	27.7	28.8	25.1	20.6	16.1	9.4
		MCDB	8.9	11.9	15.0	20.9	23.2	25.4	29.4	30.5	27.2	22.9	18.1	11.1
	2%	WB	4.8	6.2	10.5	15.1	18.8	22.6	26.6	27.5	23.9	19.1	14.2	7.3
		MCDB	6.4	8.4	13.1	19.2	22.1	25.0	28.6	29.5	26.0	21.5	16.1	9.3
	5%	WB	3.4	4.8	8.7	13.6	17.9	21.9	25.7	26.7	23.1	18.0	12.9	5.9
		MCDB	5.1	7.2	11.3	17.2	21.2	24.4	28.0	29.0	25.1	20.4	14.9	8.1
	10%	WB	1.9	3.6	7.3	12.3	16.9	21.2	25.0	25.9	22.2	17.0	11.4	4.7
		MCDB	3.6	5.8	10.1	15.8	20.4	23.8	27.3	28.6	24.3	19.6	13.7	6.8
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.5	6.9	7.3	7.5	6.6	5.1	5.4	6.8	7.3	6.6	6.2
		MCDBR	6.8	7.8	8.7	10.0	9.9	8.5	7.0	6.9	7.8	8.0	7.5	7.2
		MCWBR	5.3	5.3	5.2	4.8	4.4	3.4	3.1	3.0	3.6	4.2	5.4	5.7
	5% WB	MCDBR	6.4	7.2	7.6	8.5	7.9	6.4	5.5	5.7	5.9	7.0	6.5	6.8
		MCWBR	5.6	5.5	5.4	5.1	4.7	3.4	3.1	3.0	3.4	4.7	5.7	6.2
Clear-Sky Solar Irradiance	taub	0.433	0.485	0.562	0.607	0.621	0.694	0.662	0.577	0.508	0.500	0.481	0.433	
	taud	1.963	1.871	1.734	1.687	1.711	1.633	1.749	1.953	2.063	1.954	1.943	1.983	
	Ebn at Noon	701	716	701	701	701	650	668	718	744	694	645	666	
	Edh at Noon	137	170	214	237	236	255	225	180	153	155	139	126	
All-Sky Solar Radiation	RadAvg	2.52	3.29	4.34	5.05	5.66	5.30	4.17	4.46	4.21	3.63	2.45	2.19	
	RadStd	0.16	0.26	0.42	0.33	0.54	0.59	0.65	0.63	0.44	0.25	0.24	0.14	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+93	+50

Nomenclature: See separate page